



**CSE471: System Analysis and Design**  
**Assignment on Functional Requirements**  
**Proposed Project Title: Research Assistant Platform**

| Group No: 10, CSE471 Lab Section: 10,<br>Summer 2026 |          |                  |
|------------------------------------------------------|----------|------------------|
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# Project Overview

The Research Assistant Platform is a web-based system that supports students and researchers across the full research lifecycle - from organizing and reading papers, to identifying research gaps, to managing collaborative research projects. The system supports four user roles: Researcher, Team Member, Supervisor, and Admin. It integrates meaningful AI tooling (summarization, insight extraction, gap analysis, an interactive paper-chat assistant, and title generation) rather than adding AI for its own sake, and gathers the heavier AI features behind a Free / Premium subscription model enforced through a monthly usage quota.

## Tech Stack:

- **Language:** Python is used for both the frontend and backend layers.
- **Backend Framework:** FastAPI handles asynchronous REST API endpoints and data processing.
- **Frontend Framework:** Flask manages web page routing and renders Jinja2 templates.
- **Styling:** Tailwind CSS provides responsive, utility-first UI styling directly in HTML.
- **Database:** PostgreSQL stores relational data. It uses the pgvector extension for embeddings.
- **ORM:** SQLAlchemy manages database queries and object-relational mapping.
- **Background Processing:** Celery and Redis handle heavy PDF parsing, chunking, and embedding tasks.
- **Authentication:** JWT-based secure token management.
- **Deployment:** Hosted on Render or Railway (Backend) and Vercel or similar (Frontend).

## External APIs:

- **Google Gemini API (gemini-2.0-flash):** Powers text summarization, insight extraction, gap analysis, question generation, and paper chat.
- **SSLCommerz / Stripe API:** Processes premium subscription transactions via secure payment gateways.
- **SMTP / Gmail API:** Dispatches automated email notifications for project milestones and system deadlines.

## User Roles

1. **Researcher:** The primary user - uploads and organizes papers, takes notes, uses the AI research tools, and creates/manages research projects.
2. **Team Member:** A collaborator invited into a research project - shares tasks, papers, and discussion within that project's workspace.

3. Supervisor: Reviews a project's progress, leaves feedback on submitted work, and approves or requests revisions on milestones.
4. Admin: Platform manager - monitors overall usage, manages subscription tiers, and oversees AI request quotas and costs across users.

## Functional Requirements

| SL | Common Workflows                                                                                                                                                                                                                                                                                                                                    |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Registration and Login System: Users register by providing name, email, and password, and select a role (Researcher or Team Member); Supervisors are invited into a specific project by an existing member rather than self-registering. Once registered, users log in securely via JWT-based authentication to reach their personalized dashboard. |
| 2  | Admin Role & Access Management: An authorized Admin signs into a dedicated panel to monitor platform-wide activity - total users, AI request volume, and active subscriptions. The Admin can adjust subscription tier limits, manually resolve payment disputes, and review flagged or unusually high AI usage.                                     |

| Module 1 |                                                                                                                                                                                      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Member   | Feature Description                                                                                                                                                                  |
| 1        | Researchers upload PDF papers, which are stored in their personal library. A background worker extracts text and basic metadata (title, authors, publication year) where available   |
| 2        | Papers can be organized by topic, year, author, or research area, with custom tags/folders. The system can suggest categories using AI-generated keywords from the paper's abstract. |
| 3        | Researchers take Markdown notes linked to a specific paper, optionally tied to a page or section, timestamped and shown alongside the paper.                                         |
| 4        | Researchers can bookmark important papers or specific sections for quick access later, surfaced in a dedicated “Saved” view on the dashboard.                                        |

| Module 2 |                                                                                                                                                                                                                        |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Member   | Feature Description                                                                                                                                                                                                    |
| 1        | Generates an AI summary of an uploaded paper covering the abstract, methodology, and key findings, using the Gemini API with the extracted paper text as context.                                                      |
| 2        | Extracts a structured breakdown from a paper - contributions, advantages, limitations, and future work - instead of a single free-form summary.                                                                        |
| 3        | Given a set of papers (e.g. 5–10) selected by the user, the system synthesizes their limitations and future-work sections via the Gemini API to surface recurring unsolved problems and potential research directions. |
| 4        | Given a topic (e.g. “Traffic congestion in Dhaka”), generates candidate research questions, objectives, and hypotheses the researcher can refine and save to a project.                                                |

| Module 3 |                                                                                                                                                                            |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Member   | Feature Description                                                                                                                                                        |
| 1        | Auto-compiles a structured comparison table (Paper, Dataset, Method, Result/Accuracy) from the user's selected papers, using AI-extracted fields with manual edit support. |
| 2        | Generates citations in APA, IEEE, and ACM formats from paper metadata using deterministic rule-based templates (no AI), with one-click copy/export.                        |
| 3        | Suggests related papers from the user's library based on keyword and embedding similarity, computed dynamically rather than stored.                                        |
| 4        | Displays a side-by-side comparison of two or more papers across method, dataset, and reported results, rendered as a dynamic view over existing paper data.                |

| Module 4 |                                                                                                                                                                                            |
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| Member   | Feature Description                                                                                                                                                                        |
| 1        | Researchers create named projects (e.g. “Federated Learning for Pest Classification”) and invite Team Members and a Supervisor to collaborate.                                             |
| 2        | Project members create and assign research-specific tasks (dataset collection, model training, paper writing, etc.) with status tracking (To-Do, In Progress, Review, Done) and due dates. |
| 3        | Supervisors review submitted milestones, leave comments, and approve or request revisions; approvals are logged with the supervisor's ID and trigger notifications to the team.            |
| 4        | Tracks key project milestones (Proposal, Mid-Defense, Final Paper, Defense) on a visual timeline showing completed, pending, and upcoming stages.                                          |

| Additional Features                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Users ask natural-language questions about a specific paper (e.g. “What dataset did this paper use?”); the system retrieves relevant text chunks via embedding similarity and answers using the Gemini API. |
| Tracks each paper's read percentage, time spent, and completion status, visible from the library view.                                                                                                      |
| Sends automated email and in-app reminders ahead of upcoming research milestones and task due dates.                                                                                                        |
| A per-project discussion thread where team members and the supervisor can post updates and comments outside of formal milestone feedback.                                                                   |

A private, dated log where researchers can record daily research notes and progress.

Generates candidate paper/thesis titles from a short project description or abstract, using the Gemini API.

A meeting scheduler workspace allowing project members to request specific dates and times for consultations, which the supervisor can view, accept, or modify.

An institutional resource library listing official BRAC University thesis guidelines, formatting templates, grading rubrics, and direct links to the university's digital repository.